

September 2024 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains SEVEN questions. Answer ALL questions.

At the top of your answer scripts, please write the following:

- Your full name;
- Your class group; and
- Your class teacher's name.

Time Allowed: *2 hours*

You are supplied with: *Standard Normal and Student's t tables*

Calculators: *Calculators are NOT allowed in this examination*

1. **Total: 15 points**

- (a) An experiment consists of drawing 3 cards in succession from a well-shuffled ordinary deck of cards. Let A_i denote the event of “*obtaining king on i^{th} draw*”, for $i = 1, 2, 3$. Further, let A_i^c denote the complement of event A_i , for $i = 1, 2, 3$. State clearly in words the meaning of each of the following expressions:

- i. $\Pr(A_1 \cap A_2^c)$,
- ii. $\Pr(A_1 \cup A_2)$,
- iii. $\Pr(A_1^c \cup A_2^c)$,
- iv. $\Pr(A_1^c \cap A_2^c \cap A_3^c)$, and
- v. $\Pr((A_1 \cap A_2) \cup (A_2^c \cap A_3))$.

Your answers may be as brief as you wish but they must be precise. **(5pts)**

- (b) State the axioms of probability. Hence, prove formally that for any event A defined on sample space Ω , it holds that

$$\Pr(A^c) = 1 - \Pr(A),$$

where A^c denotes the complement of A . **(4pts)**

- (c) One bag contains 4 white balls and 2 black balls; another contains 3 white balls and 5 black balls. If one ball is drawn from each bag, find the probability that

- i. both are white,
- ii. both are black,
- iii. one is white and one is black.

You may leave your answers in terms of simplified fractions. **(6pts)**

The answer for part (a) is:

- (i) Probability of king on first draw and no king on second draw.
- (ii) Probability of king on first draw or king on second draw or king on both draws.
- (iii) Probability of no king on first draw or no king on second draw or no king on both draws.
- (iv) Probability of no king on first draw and no king on second draw and no king on third draw.
- (v) Probability of either the event of king on first draw and king on second draw or

the event of no king on second draw and king on third draw or both aforementioned events. 1 point for each of the 5 parts. (Note: if students use DeMorgan's laws to give a differently-worded but equivalent answer - i.e. for parts iii and iv – that's fine.)

The answer for part (b) is:

Let $\{A_1, A_2, \dots\}$ denote a set of events defined on sample space Ω .

Axiom 1. $\Pr(A_i) \geq 0$ for $i = 1, 2, \dots$; 1 point.

Axiom 2. $\Pr(\Omega) = 1$; and 1 point.

Axiom 3. For any events A_i and A_j , where $i \neq j$, such that $A_i \cap A_j = \emptyset$, it holds that

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i).$$

1 point.

We note that $A \cup A^c = \Omega$ and $A \cap A^c = \emptyset$. In other words, A and A^c obviously partition Ω . The required result follows immediately by observing that $\Pr(A) + \Pr(A^c) = \Pr(\Omega) = 1$, where the first equality is due to Axiom 3 and the final equality is due to Axiom 2. 1 point.

The answer for part (c) is:

Let W_1 denote event “white ball from first bag” and W_2 denote event “white ball from second bag”. Then,

- (i) $\Pr(W_1 \cap W_2) = \Pr(W_1)\Pr(W_2|W_1) = \Pr(W_1)\Pr(W_2) = \left(\frac{4}{4+2}\right)\left(\frac{3}{3+5}\right) = \frac{1}{4}$;
- (ii) $\Pr(W_1^c \cap W_2^c) = \Pr(W_1^c)\Pr(W_2^c|W_1^c) = \Pr(W_1^c)\Pr(W_2^c) = \left(\frac{2}{4+2}\right)\left(\frac{5}{3+5}\right) = \frac{5}{24}$;
- (iii) We have $1 - \Pr(W_1 \cap W_2) - \Pr(W_1^c \cap W_2^c) = 1 - \frac{1}{4} - \frac{5}{24} = \frac{13}{24}$.

2 points for each of the 3 parts.

2. Total: 15 points

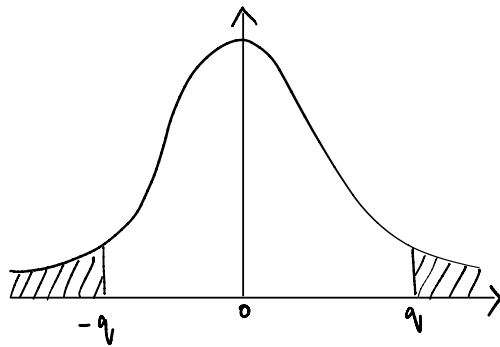
- (a) Pakistani athlete, Arshad Nadeem, set an Olympic record with a javelin throw of 92.97 metres (305.0 feet) at the 2024 Summer Olympics in Paris. Suppose we were to model the length of any one of his throws, say X , as a continuous random variable with a well-defined probability density function, say f_X . Such a model, of course, enforces that the probability that X takes any specific real value is zero. *But the javelin has to land somewhere!* Does that mean our model is useless? Explain your answer. **(5pts)**

- (b) Find the value of constant c such that the function

$$f_X(x) = \begin{cases} cx^2 & 0 < x < 3 \\ 0 & \text{otherwise} \end{cases}$$

is a density function for random variable X . Hence, work out $\Pr(1 < X \leq 2)$. Evaluate the cumulative distribution function of X , say $F_X(x)$, and use $F_X(x)$ to confirm your previous answer. **(5pts)**

- (c) A rough sketch of Student's t distribution with 9 degrees of freedom is below.



Find the value of point $q \in \mathbb{R}^+$ (as marked on the horizontal axis) for which:

- the shaded area on the right side of the graph is 0.05,
- the total shaded area on the graph is 0.05,
- the total unshaded area on the graph is 0.99,
- the shaded area on the left of the graph is 0.01, and
- the area on the graph to the left of point q is 0.90.

(5pts)

For part (a):

The apparent paradox is resolved by considering precisely what it means to measure something on a continuous scale. Suppose when we measure my throw it comes to 513cm (I'm not an athlete!). If our tape measure is accurate to within 0.5cm, our measurement actually indicates that the throw fell somewhere in the interval [512.5, 513.5]. This interval has some non-zero probability associated with it. No matter how much we improve our measuring device, there will always be limitations to its accuracy. Thus, any measurement on a continuous scale effectively yields an interval rather than a single value. **2 points – students don't have to say exactly these words but any sensible explanation that conveys the key message will earn marks.**

If X is a continuous random variable with density f_X , and $a, b \in \mathbb{R}$ with $a \leq b$, then

$$\Pr(a < X \leq b) = \int_a^b f_X(x) \, dx,$$

which will clearly be 0 when $a = b$ but certainly has the potential to be non-zero for $a < b$. In other words, probability is not associated with the values of the density function but with the area beneath the curve that the density function defines. In order to work out the probability that a random variable X takes a value between a and b , we work out the area above the x -axis, beneath the density, and between the lines $x = a$ and $x = b$. **2 points – again, students don't have to give my exact expressions but will earn these marks for supporting their answer with some algebra** So, no the model is not useless (i.e. at least not on the basis of the argument given in the question). **1 point**

For part (b:)

Since $f_X(x)$ is a valid density, $\int_{-\infty}^{\infty} f_X(x) \, dx = \int_0^3 cx^2 \, dx = 9c = 1$. Thus, we have $c = 1/9$. Further, $P(1 < X \leq 2) = \int_1^2 (1/9)x^2 \, dx = 7/27$. Next, the CDF is

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^3/27 & 0 \leq x < 3 \\ 1 & x \geq 3. \end{cases}$$

We thereby confirm that $P(1 < X \leq 2) = F_X(2) - F_X(1) = 7/27$.

1 point for normalising constant, 1 point for PDF, 1 point for obtaining probability, 1 point for CDF, and 1 point for confirmation of probability using CDF.

For part (c):

i. 1.83,

ii. 2.26,

iii. 3.25,

iv. 2.82,

v. 1.38. 1 point for each of the 5 parts.

3. Total: 10 points

(a) Consider an Exponential(λ) random variable, X , whose density is given by

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$. Also consider a Gamma(n, λ) random variable, Y , with density

$$f_Y(y) = \begin{cases} e^{-\lambda y} y^{n-1} \lambda^n / (n-1)!, & y \geq 0 \\ 0, & \text{otherwise,} \end{cases}$$

where n is a positive integer and the exclamation point denotes a factorial. Derive the moment-generating function of X , and also show that the moment-generating function of Y is given by

$$M_Y(t) = \left(1 - \frac{t}{\lambda}\right)^{-n}$$

for $t < \lambda$.

(6pts)

(b) Your colleague states that random variable $Y \sim \text{Gamma}(n, \lambda)$ may be represented as $Y = \sum_{i=1}^n X_i$, where $X_i \stackrel{iid}{\sim} \text{Exponential}(\lambda)$ for $i = 1, \dots, n$. Prove this statement using the moment-generating functions of Y and $X_1, \dots, X_n$. **(4pts)**

$$Q3(a) \quad M_X(t) := E[e^{tx}] = \int_{-\infty}^{\infty} e^{tx} \lambda e^{-\lambda x} dx$$

1

$$= \int_{-\infty}^{\infty} \lambda \exp(-(1-t)) dx$$

$$= \frac{\lambda}{\lambda-t} \int_{-\infty}^{\infty} (\lambda-t) \exp(-(1-t)) x dx$$

$$= \frac{\lambda}{\lambda-t} \quad \text{for } t < \lambda,$$

1

since the integral is of the $\text{Exp}(\lambda-t)$ PDF over its entire support.

1

$$M_Y(t) := E[e^{ty}] = \int_{-\infty}^{\infty} \exp(ty) \exp(-\lambda y) y^{n-1} \lambda^n / (n-1)! dy$$

1

$$= \int_{-\infty}^{\infty} \exp(-(1-t)y) y^{n-1} \lambda^n / (n-1)! dy$$

$$= \left(\frac{\lambda}{\lambda - t} \right)^n \int_{-\infty}^{\infty} \exp(-(\lambda - t) y) y^{n-1} (\lambda - t)^n / (n-1)! dy$$

$$(eq. 1) = \left[\lambda / (\lambda - t) \right]^n \text{ for } t < \lambda, \quad 1$$

since the integral is of the $\text{Gamma}(n, (\lambda - t))$ PDF over its entire support. 1

The required result follows by dividing numerator and denominator of the ratio in square brackets in (eq. 1) by λ , and then considering its reciprocal.

Q3(b) Suppose $Y = X_1 + \dots + X_n$, where $X_i \stackrel{i.i.d.}{\sim} \text{Exp}(\lambda)$ for $i = 1, \dots, n$.

$$\begin{aligned} M_Y(t) &= E[e^{tY}] = E[e^{t(X_1 + \dots + X_n)}] \stackrel{i.i.d.}{=} \left(E[e^{tX_1}] \right)^n \\ &= \left(M_X(t) \right)^n = \left(\frac{\lambda}{\lambda - t} \right)^n \text{ for } t < \lambda. \end{aligned} \quad \left. \vphantom{\begin{aligned} M_Y(t) &= E[e^{tY}] = E[e^{t(X_1 + \dots + X_n)}] \stackrel{i.i.d.}{=} \left(E[e^{tX_1}] \right)^n \\ &= \left(M_X(t) \right)^n = \left(\frac{\lambda}{\lambda - t} \right)^n \text{ for } t < \lambda. \end{aligned}} \right\} 2$$

Indeed, by the uniqueness property of moment-generating functions, it is immediate that $Y \sim \text{Gamma}(n, \lambda)$ 1

Award the final mark to anyone who notices that, in fact, saying " $X \sim \text{Exp}(\lambda)$ " is equivalent to saying " $X \sim \text{Gamma}(1, \lambda)$." 1

4. **Total: 15 points**

- (a) Suppose $\hat{\mu}$ is an unbiased estimator of μ . You are informed that the mean-square error (MSE) of $\hat{\mu}$ is 30. What is $E(\hat{\mu}^2)$ when $\mu = 6$? **(5pts)**
- (b) Let $X_1, \dots, X_n$ denote a random sample from an Exponential($1/\theta$) distribution with density

$$f_X(x) = \begin{cases} \frac{1}{\theta} e^{-x/\theta} & x \geq 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $\theta > 0$. Derive the MSE for each of the following estimators of θ :

- i. $\hat{\theta}_1 = X_1$,
- ii. $\hat{\theta}_2 = (X_1 + X_2)/2$,
- iii. $\hat{\theta}_3 = (X_1 + 2X_2)/3$,
- iv. $\hat{\theta}_4 = (X_1 + \dots + X_n)/n$, and
- v. $\hat{\theta}_5 = 5$.

You may use (without proof) the result that for an Exponential($1/\theta$) random variable X , we have $E(X) = \theta$ and $\text{Var}(X) = \theta^2$. **(10pts)**

$$\begin{aligned}
 (a) \quad \text{MSE}(\hat{\mu}) &= [E(\hat{\mu}) - \mu]^2 + \text{Var}(\hat{\mu}) & 1 \\
 &= 0 + E(\hat{\mu}^2) - (E(\hat{\mu}))^2 \text{ due to unbiasedness} & 2
 \end{aligned}$$

$$\Rightarrow E(\hat{\mu}^2) = 36 + 30 = 66 \quad 2$$

(b) $\hat{\theta}_1, \dots, \hat{\theta}_4$ are unbiased. $\hat{\theta}_5$ has zero variance.

$$\text{Hence, } \text{MSE}(\hat{\theta}_1) = \text{Var}(X_1) = \underline{\underline{\theta^2}} \quad 2$$

$$\text{MSE}(\hat{\theta}_2) \stackrel{||\Delta}{=} (1/2)^2 [\text{Var}(X_1) + \text{Var}(X_2)] = \underline{\underline{\theta^2/2}} \quad 2$$

$$\text{MSE}(\hat{\theta}_3) \stackrel{||\Delta}{=} (4/3)^2 \text{Var}(X_1) + (2/3)^2 \text{Var}(X_2)$$

$$= (16/9) \theta^2 + (4/9) \theta^2$$

$$= \underline{\underline{(5/9) \theta^2}} \quad 2$$

$$\text{MSE}(\hat{\theta}_4) \stackrel{||\Delta}{=} n \text{Var}(X_1) / n^2 = \underline{\underline{\theta^2/n}} \quad 2$$

$$\text{MSE}(\hat{\theta}_5) = \underline{\underline{(5-\theta)^2}} \quad 2$$

5. **Total: 15 points**

- (a) Suppose $X_i \stackrel{iid}{\sim} \text{Geometric}(\pi)$ for $i = 1, \dots, n$, where π is unknown. In other words, the probability mass function of each of the X_i variables is given by

$$\Pr(X_i = x) = \begin{cases} (1 - \pi)^{x-1} \pi & x = 1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Derive the maximum likelihood estimator for π given a sample $\mathbf{x} = (x_1, \dots, x_n)'$.
(You do **not** need to verify the second-order condition.) **(9pts)**

- (b) Consider a continuous random variable X which is uniformly distributed over the interval $[-\alpha, \alpha]$ for some $\alpha \in \mathbb{R}^+$. In other words, the density is $f_X(x) = 1/2\alpha$ for $x \in [-\alpha, \alpha]$, and 0 otherwise. Suppose we have a random sample of size n , denoted as $(x_1, \dots, x_n)$, from this distribution. Your colleague guesses that the sample mean $(1/n) \sum_{i=1}^n x_i$ yields the maximum likelihood (ML) estimate of α . Explain in words (without using equations) why your colleague's guess is likely to be incorrect. What do *you* guess will be the ML estimate of α ? **(6pts)**
-

Q5. The likelihood function is

$$\mathcal{L}(\pi; x) = P(X_1 = x_1, \dots, X_n = x_n; \pi) \stackrel{\text{IID}}{=} \prod_{i=1}^n P(X_i = x_i)$$

$$= \prod_{i=1}^n (1-\pi)^{x_i-1} \pi = (1-\pi)^{\left(\sum_{i=1}^n x_i - n\right)} \pi^n$$

where $X := (X_1, \dots, X_n)'$.

The log-likelihood function is

$$\ell(\pi; x) = \log \mathcal{L}(\pi; x) = \left[\sum_{i=1}^n x_i - n \right] \log(1-\pi) + n \log \pi$$

The score function is

$$\mathcal{A}(\pi; x) = \left(\frac{\sum_{i=1}^n x_i - n}{1-\pi} \right) + \frac{n}{\pi}.$$

Solving the equation $\mathcal{A}(\hat{\pi}_{MLE}; x) = 0$ gives us $\hat{\pi}_{MLE} = 1/\bar{x}$.

For part (a) – 2 points for likelihood; 2 for log-likelihood; 2 for first derivative; 2 for final answer; award the final 1 mark if student remembers to explicitly put a hat on π when setting score to zero.

For part (b):

The colleague's guess is likely to be incorrect because the sample mean is a statistic that typically helps estimate parameters that characterise central tendency of a random variable. In contrast, the parameter α here is extremal (and the population mean in this case is just zero). **3 points.** Given the nature of the problem, it would make more sense to consider the maximum over the set $\{|x_1|, \dots, |x_n|\}$ as an estimator for α . **3 points.**

6. Total: 20 points

- (a) A machine previously produced washers having a mean thickness of 0.050 inches. To determine whether the machine is currently in proper working order, a random sample of 10 washers is chosen for which the mean thickness is 0.053 inches and the standard deviation is 0.003 inches. In other words, using conventional notation, let us denote

$$\bar{x} = (x_1 + \dots + x_{10})/10 = 0.053$$

and

$$\hat{\sigma} = \sqrt{(x_1^2 + \dots + x_{10}^2)/10} = 0.003.$$

Your goal is to test the hypothesis that the machine is currently in proper working order, and your colleague proposes that you use the test statistic

$$\frac{\sqrt{9}(\bar{x} - 0.050)}{\hat{\sigma}},$$

along with Student's t_9 distribution, to achieve this goal.

Provide, using your colleague's suggestion, a detailed step-by-step formulation of a test of the hypothesis that the machine is in proper working order at both 5% and 1% significance levels. Also give, with explicit reference to the statistical tables provided, your best approximation of the p -value of the test. **(15pts)**

- (b) Explain what are Type I and Type II errors in statistical hypothesis testing in any way you consider appropriate. (You may use algebra, tables/graphs, real-life examples, or any combination of these methods. Marks will be awarded for evidence of knowledge and clarity of exposition.) **(5pts)**

Assumptions let x_i denote thickness of washer i
for $i=1, \dots, n$. Then, suppose $x_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$
for $i=1, \dots, n$. 2

Hypotheses $H_0: \mu = \mu_0 := 0.050$ 1
 $H_1: \mu \neq \mu_0$

Test Statistic $V = \frac{\sqrt{n-1}(\bar{x} - \mu_0)}{\hat{\sigma}}$ where $\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$ 2

Dist. of V under H_0 $V \sim t_{n-1}$ 1

Significance levels $\alpha \in \{0.05, 0.01\}$ 1

Critical values $t_{n-1}^{\alpha/2} \in \{2.26, 3.25\}$

2

Decision rule Reject H_0 iff $|V| > t_{n-1}^{\alpha/2}$

1

Computed value of V $\frac{(0.053 - 0.050) \sqrt{10-1}}{0.003} = 3$

1

Test Result Since $2.26 < 3 < 3.25$, we reject H_0 at the 5% but not the 1% level.

2

Interpretation

There is moderate evidence against the null.
It would be advisable to check the machine or at least take another / bigger sample.

2

P-value

$$p = P(V > 3) + P(V \leq -3)$$

$$\in [0.01, 0.02]$$

According to row 9 of the table in the last 2 columns. Since $2.82 < 3 < 3.25$, we know that $P(V > 3)$ is between 0.005 and 0.010 and we double this due to symmetry.

NO MARKS ASSIGNED FOR P-VALUE

For part (c): **Textbook.** So long as the student demonstrates knowledge of the concept and is able to convey their point clearly, they can earn full marks.

7. Total: 10 points

Suppose $X_t \stackrel{iid}{\sim} (\mu, \sigma_X^2)$ for $t = 1, \dots, T$ with $0 < \sigma_X^2 < \infty$ and some $T \in \mathbb{N}$.

- (a) State Chebyshev's law of large numbers, and then prove it. In your answer, you may use (without proof) Chebyshev's inequality, which states that for any random variable X with finite variance and any $\epsilon \in \mathbb{R}^+$,

$$\Pr(|X - \mathbb{E}(X)| > \epsilon) \leq \frac{\text{Var}(X)}{\epsilon^2}.$$

(7pts)

- (b) State the (Lindeberg-Lévy) central limit theorem (CLT).
(You do **not** need to prove it.)

(3pts)

END OF EXAM PAPER

(a) PROPOSITION. Given random variables $X_t \stackrel{i.i.d.}{\sim} (\mu, \sigma_x^2)$ for $t=1, \dots, T$, with $T \in \mathbb{N}$ and $0 < \sigma_x^2 < \infty$, it holds that for any $\varepsilon > 0$,

$$\lim_{T \rightarrow \infty} P(|\bar{X}_T - \mu| > \varepsilon) = 0,$$

$$\text{where } \bar{X}_T := \frac{1}{T} \sum_{t=1}^T X_t.$$

3
□

PROOF. By linearity of the expectation operator,

$$E(\bar{X}_T) = E\left(\frac{1}{T} \sum_{t=1}^T X_t\right) = \frac{1}{T} \sum_{t=1}^T E(X_t) = (T\mu)/T = \mu.$$

$$\text{Further, } \text{Var}(\bar{X}_T) = \frac{1}{T^2} \text{Var}\left(\sum_{t=1}^T X_t\right) = \frac{1}{T^2} \sum_{t=1}^T \text{Var}(X_t)$$

$$= \frac{\sigma_x^2}{T}.$$

It follows that $\text{MSE}(\bar{X}_T) := \text{Bias}^2(\bar{X}_T) + \text{Var}(\bar{X}_T) = \sigma_x^2/T$,
so that $\lim_{T \rightarrow \infty} \text{MSE}(\bar{X}_T) = 0$.

Finally, we have by Markov's inequality that for any $\varepsilon > 0$,
$$\mathbb{P}(|\bar{X}_T - \mu| > \varepsilon) = \mathbb{P}((\bar{X}_T - \mu)^2 > \varepsilon^2) \leq \frac{E[(\bar{X}_T - \mu)^2]}{\varepsilon^2} = \frac{\sigma_x^2}{T\varepsilon^2},$$

from which the required result is immediate. 4 

(b) PROPOSITION. Given random variables $X_t \stackrel{i.i.d.}{\sim} (\mu, \sigma_x^2)$ for $t=1, \dots, T$, with $T \in \mathbb{N}$ and $0 < \sigma_x^2 < \infty$, it holds that

$$\sqrt{T} \left(\frac{\bar{X}_T - \mu}{\sigma_x} \right) \xrightarrow{d} N(0, 1) \text{ as } T \rightarrow \infty$$

where $\bar{X}_T := \frac{1}{T} \sum_{t=1}^T X_t$.

3
□

TABLE FD.1 Observations on Income and Education, 20 Observations

I = Observation,
 Y = Income.

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z \leq z]$

[illegible]

TABLE G.2 Percentiles of the Student's t Distribution. Table Entry Is x Such that $\text{Prob}[t_n \leq x] = P$

n	.750	.900	.950	.975	.990	.995
1	1.000	3.078	6.314	12.706	31.821	63.657
2	.816	1.886	2.920	4.303	6.965	9.925
3	.765	1.638	2.353	3.182	4.541	5.841
4	.741	1.533	2.132	2.776	3.747	4.604
5	.727	1.476	2.015	2.571	3.365	4.032
6	.718	1.440	1.943	2.447	3.143	3.707
7	.711	1.415	1.895	2.365	2.998	3.499
8	.706	1.397	1.860	2.306	2.896	3.355
9	.703	1.383	1.833	2.262	2.821	3.250
10	.700	1.372	1.812	2.228	2.764	3.169
11	.697	1.363	1.796	2.201	2.718	3.106
12	.695	1.356	1.782	2.179	2.681	3.055
13	.694	1.350	1.771	2.160	2.650	3.012
14	.692	1.345	1.761	2.145	2.624	2.977
15	.691	1.341	1.753	2.131	2.602	2.947
16	.690	1.337	1.746	2.120	2.583	2.921
17	.689	1.333	1.740	2.110	2.567	2.898
18	.688	1.330	1.734	2.101	2.552	2.878
19	.688	1.328	1.729	2.093	2.539	2.861
20	.687	1.325	1.725	2.086	2.528	2.845
21	.686	1.323	1.721	2.080	2.518	2.831
22	.686	1.321	1.717	2.074	2.508	2.819
23	.685	1.319	1.714	2.069	2.500	2.807
24	.685	1.318	1.711	2.064	2.492	2.797
25	.684	1.316	1.708	2.060	2.485	2.787
26	.684	1.315	1.706	2.056	2.479	2.779
27	.684	1.314	1.703	2.052	2.473	2.771
28	.683	1.313	1.701	2.048	2.467	2.763
29	.683	1.311	1.699	2.045	2.462	2.756
30	.683	1.310	1.697	2.042	2.457	2.750
35	.682	1.306	1.690	2.030	2.438	2.724
40	.681	1.303	1.684	2.021	2.423	2.704
45	.680	1.301	1.679	2.014	2.412	2.690
50	.679	1.299	1.676	2.009	2.403	2.678
60	.679	1.296	1.671	2.000	2.390	2.660
70	.678	1.294	1.667	1.994	2.381	2.648
80	.678	1.292	1.664	1.990	2.374	2.639
90	.677	1.291	1.662	1.987	2.368	2.632
100	.677	1.290	1.660	1.984	2.364	2.626
∞	.674	1.282	1.645	1.960	2.326	2.576